Specifications Of Software Systems

CEN 3016 921184

Fall 2026 - Term

Course Information

Credit Hours: 3.0
Course Description: Specifications Of Software Systems
Looks at specifications that have well defined semantics. Covers classes of specification models, including algebraic, petri-nets and model-theoretic approaches. Prerequisite: CEN3011C

Class Information

Class modality: Face to Face

Class Schedule: Tuesday and Thursday (TR), 1:00-2:15 PM

Location: Classroom B103

Faculty Information

Oscar Somarriba
Email: oscar.somarriba@keiseruniversity.edu
Phone: +505-88825648

Credentials
PhD

Office Hours

TR 10 AM-12:30 PM (meeting previously agreed by WhatsApp)-LIBRARY

Program Mission

The Bachelor of Science in Information Technology program is designed to develop foundational knowledge of Information Technology practices and procedures to support continuous organizational improvement activities. The program prepares students to become networking or cybersecurity practitioners in a variety of technical and non-technical work roles. Focus is paid to develop business communication, empathy, and interaction skills to promote organizational culture.

Program Goal(s)

1. Students will develop foundational knowledge of Information Technology practices and procedures to support continuous organizational improvement activities.

2. Students will assess issues affecting technological environments and human behaviors to determine best practices to solve complex problems.

3. (Networking Concentration): Students will develop proficiency in new and existing infrastructures to provide available and secure networks.

4. (Cybersecurity Concentration): Students will survey management controls to govern, identify, protect, detect, respond, and recover from cyber risk.

Program - Student Learning Objectives/Outcomes

- Objective/Outcome: 1. Students will support on-premises or remote technology environments to maintain and advance organizational needs.
- Objective/Outcome: 2. Students will develop and provide training about new technologies and best practices used by anyone accessing organizational assets.

- Objective/Outcome: 3. Students will document best practices and techniques to illustrate complex configurations.
- Objective/Outcome: 4. Students will research and document regulatory/contractual compliance requirements, artificial intelligence (AI) cyber risk management and Internet of Things (IoT) environments.

Course Learning Objectives

- Upon completion of the course, students are able to:
- 1. Understand the importance for specifying a software system
- 2. Specify software systems with UML
- 3. Specify simple concurrent systems using petri-nets
- 4. Understand temporal logic
- 5. Augment UML designs with OCL

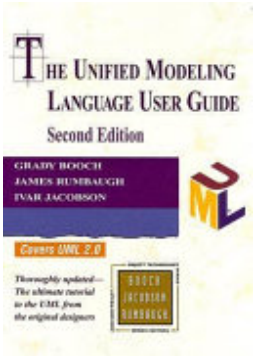
Grading and Evaluation Methods

Evaluation	Percent of Total Grade	Due Date
Group Course Project	20%	December 04, 2026
Mid-term examination	15%	Oct 15, 2026
Assignments (designs, cases, problems, other written assignments), participation	35%	December 04, 2026
Quizzes	10%	December 01, 2026
Final Exam	15%	December 9-16, 2026
Post Test	5%	December 04, 2026
Total Percentage:	100%	

Grading Scale

Letter Grade	Numeric Grade
A	90.00-100.00%
B	80.00-89.99%
C	70.00-79.99%
D	65.00-69.99%
F	Up to 64.99%

Required Textbook

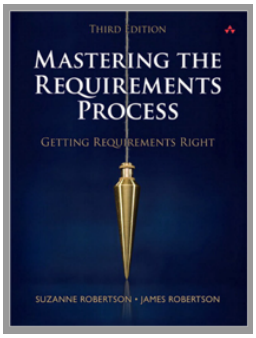


The Unified Modeling Language User Guide
Authors: Grady Booch, James Rumbaugh, Ivar Jacobson
Publisher: Addison-Wesley Professional
Publication Date: 2005
Edition: 2nd ed.

Recommended Resources



Clean Architecture A Craftsman's Guide to Software Structure and Design
ISBN: 0-13-449416-4
Authors: Robert C. Martin
Publisher: Prentice Hall
Publication Date: 2017
Edition: 1st. edition



Mastering the Requirements Process
ISBN: 978-0137969500
Authors: James Robertson, Suzanne Robertson, and Adrian Reed
Publisher: Addison-Wesley Professional
Publication Date: 2024
Edition: 4th. edition

Topical Outline/Course Assignments/Calendar

Master Syllabus

WEEK 1

Learning Objective(s): · 1

Assigned Reading: · Textbook - Chapters 1, 2,3

Suggested Learning

Activities: · Describe simple situations using propositional logic · VIDEO: http://www.criticalthinkeracademy.com/introduction1.html

Suggested Homework: · Download and install starUML · Use of sets · Use of predicate logic

Evaluation: · Pre-test, Homework, laboratory activities and Week 1 test

WEEK 2

Learning Objective(s): · 2

Assigned Reading: · Textbook – Chapters 4-8

Suggested Learning

Activities: · Specify a simple concurrent systems using Petri-nets · Petri-net simulator link: http://torguet.free.fr/java/Petri.html · Specify a simple concurrent system using TLA+

Suggested Homework: · Petri-net exercises · TLA+ exercise

Evaluation: · Homework, laboratory activities and mid-term examination

WEEK 3

Learning Objective(s): · 2,3

Assigned Reading: · Textbook – Chapter 9-15

Suggested Learning

Activities: · Continuance of TLA+ specification

Suggested Homework: · Chapter 5 Review Questions

Evaluation: · Homework, laboratory activities and Week 3 test

WEEK 4

Learning Objective(s): · 2,4,5

Assigned Reading: · Textbook – Chapter 17-25

Suggested Learning

Activities: · Temporal logic Link: http://www.inf.unibz.it/~artale/FM/slide3.pdf · Revisit UML diagrams – Emphasize UML as a semi-formal language for specification.

Suggested Homework: · UML/OCL exercises

Evaluation: · Homework, laboratory activities, final examination and post-test

Topical Outline/Course Assignments/Calendar (Detailed Syllabus)

Week 1, day 1, 1:00pm to 2:15pm Introductions, review course syllabus, review class policies, procedures and expectations, take pre-test

Week 1, day 2, 1:00pm to 2:15pm Lecture on Chapters 1 & 2 (book by Robertson et al.): Requirements Fundamentals & Your Requirements Arena, classwork1 (Essay on Requirement Engineering), Hw 1:UML Practice, Review question 1 (RQ1)

Week 2, day 1, 1:00pm to 2:15pm Lecture on Chapters 3 & 4 (book by Robertson et al.): Understanding the real problem & The Value of Solving the Problem, classwork2

Week 2, day 2, 1:00pm to 2:15pm Hw2 (Assignment: Restaurant Case), RQ2

Week 3, day 1, (September 15, 2026) Central American Independent Holiday.

Week 3, day 3, 1:00pm to 2:15pm Lecture on chapters 1, 2 & 3 (book by R.C Martin): What is Design and Architecture, A Tale of Two Values, and Paradigm Overview, classwork3 (Essay on Clean Architecture), RQ3, Hw3

Week 4, day 1 (September 22, 2026) Independent Fest (Holiday)

Week 4, day 2, 1:00pm to 2:15pm Lecture on Chapters 3-6 (book by R.C Martin), classsswork4, Hw4, RQ4

Week 5, day 1, 1:00pm to 2:15pm Lecture on Chapters 7-9 (book by R.C Martin), classwork5 (Discuss the SRP, OCP, LSP SOLID principles)

Week 5, day 2, 1:00pm to 2:15pm Hw5, RQ5, quiz1 (Chapters 1-6, book by R.C. Martin)

Week 6, day 1, 1:00pm to 2:15pm Lecture on Chapters 10-11 (book by R.C Martin), classwork6 (Discuss the ISP, DIP SOLID principles)

Week 6, day 2, 1:00pm to 2:15pm Hw6, RQ6

Week 7, day 1, 1:00pm to 2:15pm Lecture on Chapters 12-14 (book by R.C Martin), classwork7(Discuss on component cohesion and coupling)

Week 7, day 2, 1:00pm to 2:15pm Hw7, RQ7, midterm examination

Week 8, day 1, 1:00pm to 2:15pm Lecture on Chapters 15-17 (book by R.C Martin), classwork8 (Discuss entities and uses cases)

Week 8, day 2, 1:00pm to 2:15pm Hw8, RQ8

Week 9, day 1, 1:00pm to 2:15pm Lecture on Chapters 18-20 (book by R.C Martin), classwork9 (Discuss Boundaries, Policy and Level, Business Rules)

Week 9, day 2,1:00pm to 2:15pm Hw9, Guidelines of the class or course project, RQ9

Week 10, day 1, 1:00pm to 2:15pm Lecture on Chapters 18-20 (book by R.C Martin), classwork10 (Discuss entities and use cases)

Week10, day 2, 1:00pm to 2:15pm Hw10, project registration, RQ10, quiz2 (Chapters 7-20)

Week 11, day 1, 1:00pm to 2:15pm Lecture on Chapters 21-23 (book by R.C Martin), classwork11 (Discuss Clean Architecture)

Week11, day 2, 1:00pm to 2:15pm Hw11, project progress report, RQ11

Week 12, day 1, 21:00pm to 2:15pm Lecture on Chapters 24-26 (book by R.C Martin), classwor12 (Discuss about the Layers and boundaries in clean architecture)

Week12, day 2, 1:00pm to 2:15pm Hw12, class project discussion, RQ12

Week 13, day 1, 1:00pm to 2:15pm Lecture on chapters 27-30 book by R.C Martin), classwork13 (Discuss process and threads), Hw13, Final project reports

Week 13, day 2, Thanksgiving Holiday

Week 14, day 1, 1:00pm to 2:15pm Lecture on chapters 31-33 (book by R.C Martin), Hw14

Week14, day 2, 1:00pm to 2:15pm Post Test , Final Presentations of class project1

Week 15, Final Presentations of class project2

Week 16, Final exam

Course Guidelines and Policies

Academic Integrity

Students are expected to maintain the highest standards of academic conduct, professional honesty, and personal integrity. Plagiarism, cheating, and other misconduct are serious violations and will not be tolerated, and may result in academic penalties, including suspension or dismissal.

Participation

Participation is a basic requirement for an effective learning community. Students' participation will be assessed and reflected in academic advising sessions and/or student academic progress reports. Class/course participation is based on the following criteria: arriving to class on-time (Face to Face); participation in online chats and engagement in relevant course discussions (Face to Face and On-line); attention during lectures/discussions/debates; respectful listening when someone else is speaking or providing written responses to content topics/assignment. Respect for your leaning environment, without distractions such as texting, side-bar conversations, checking phones or emails during classroom scheduled times and/or established synchronous learning times is an essential to achieving academic success.

Missed Tests/Quizzes

Makeup exams will be allowed only with pre-approval of the instructor or with an acceptable, documented reason. Acceptable reasons for makeup exams include severe illness, family emergency or other unavoidable events. Exam format for makeup exams may be different than the original exam but the content for the exam will not change.

Late Assignments

Assignments are due at the start of class on or prior to the day noted (or as designated by the instructor). Late assignments without penalty will be accepted only in cases of emergency. Students should discuss turning in late work directly with the instructor and in advance of the due date whenever possible. Late assignments will not be accepted, if the assignment has already been graded and returned to the student. Please note that all work must be submitted by the last day of the class.

Civility/Professionalism

This class is a community of learners, which means we will depend upon each other for support and information. In order to learn, we must be open to the views of people different than ourselves. Please honor the uniqueness of your classmates and appreciate the opportunity we have to learn from one another. Please respect each other’s opinions and refrain from personal attacks or demeaning comments of any kind.

It is of the utmost importance to communicate with courtesy and professionalism. Professional courtesy includes respecting other’s opinions, being courteous and respectful, and working together in the spirit of cooperation.

Specific Uses of Artificial Intelligence (AI)

There are numerous types of Artificial Intelligence in use today, and it is expected that more will be forthcoming with regularity. For purposes of the University’s position, AI is defined as the advancement in technologies that involves many sub-categories of learning or machine-based intelligence, and those that may yet emerge. The position of the university on AI is

therefore evolving and subject to change. The university recognizes the range of possibilities for AI to be used positively to enhance learning, as well as opportunities for students to use AI negatively to produce actual work, assignments, projects, and examination responses, in ways not intended by the course instructions or by the professor. The university also recognizes that there are instances in which AI can be used as a teaching tool by professors, for pre-writing, editing, for knowledge, comprehension, and application-based learning, as well as for brainstorming, developing ideas, and many other possibilities. It is critical for the university to educate and train students in the ability to think critically to perform in their fields of study, requiring originality, competence, creativity, and independence of thought. It is the university's position that all student work should be original, be developed by the student through his or her own critical thought, and that the integrity of academic work in the advancement of knowledge be maintained through such originality in all student work, including online work, in class work, laboratory work, assignments, discussions, projects, and exams. As such, submissions of any work written by, or developed using AI, and/or outside the boundaries of the assignment as described in the syllabus or other formal course instructions, without the professor's explicit knowledge and approval, will be considered a violation of the academic honesty policy. The student must disclose any and all uses of AI in their work, and it is the student's responsibility to provide proof of originality and clear use of appropriate sources, including when requested by the professor for further clarification after submission of work.

University and Program Policies

Students are expected to abide by the policies set forth in the University Undergraduate and Graduate Catalogs and the Student Program Handbook/Manuals. The University Catalogs are available electronically at http://www.keiseruniversity.edu/catalog/. The Program Student Handbook/Manual is available electronically at the direction of your instructor.

Disability Accommodations

In compliance with the Americans with Disabilities Act (ADA), students who require reasonable accommodations due to a disability to properly execute coursework must complete the application process and receive approval from the review committee. The first step is to consult with the Campus President or Dean of Academic Affairs.